Q.1 Which of the following is a correct update formula for the Gauss-Seidel iteration in solving the finite difference equations for the Laplace PDE?

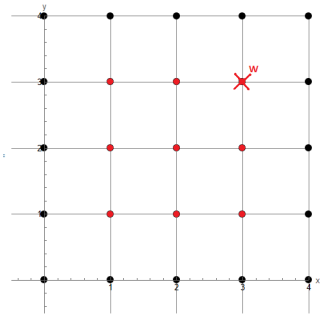
A. u_{i,j}^{(k+1)} = {1}{4} (u_{i+1,j}^{(k)} + u_{i-1,j}^{(k)} + u_{i,j+1}^{(k+1)} + u_{i,j-1}^{(k+1)})

B. u_{i,j}^{(k+1)} = {1}{4} (u_{i+1,j}^{(k)} + u_{i-1,j}^{(k+1)} + u_{i,j+1}^{(k)} + u_{i,j-1}^{(k)})

C. u_{i,j}^{(k+1)} = {1}{4} (u_{i+1,j}^{(k+1)} + u_{i-1,j}^{(k)} + u_{i,j+1}^{(k)} + u_{i,j-1}^{(k)})

D. u_{i,j}^{(k+1)} = {1}{4} (u_{i+1,j}^{(k+1)} + u_{i-1,j}^{(k+1)} + u_{i,j+1}^{(k)} + u_{i,j-1}^{(k)})

Q.2 The subscript (i,j) of w in the grid (see Fig. below) in Finite Difference Scheme for the solution of a PDE, are_____



A. (1,1)

B. (1,3)

C. (3,1)

D. (3,3)

Q.3 .

Consider the mesh point $w_l = w_{i,j}$ *of the solution domain*

where l denotes the labeling to the internal grid points *and* $l =$

A.

$i + (n-1-j)(m-1)$

B.

$i + (m+1+j)(n-1)$

C.

$i + (m-1-j)(n-1)$

D.

$j + (m-1-i)(n-1)$

Q.4 The boundary conditions u(0,y)=0 and u(1,y)=y for the differential equation \frac{\partial ^ 2 u}{\partial x ^2} + \frac{\partial ^ 2 u}{\partial y ^2} = 0 shows that the length of x is

A. 1

B. 2

C. 0.5

D. 0

Q.5 The PDE \frac{\partial ^ 2 u}{\partial x ^2} + \frac{\partial ^ 2 u}{\partial y ^2} = 0 is _____

A. Parabolic

B. Non-linear

C. Hyperbolic

D. Elliptic

Q.6 Which form of the finite difference approximation is typically used for solving the Poisson equation using the Gauss-Seidel method?

A. Fourth-order central difference

B. First-order forward difference

C. First-order backward difference

D. Second-order central difference

Q.7 Gauss-Seidel method is also known as..... Method.

A. Successive displacement

B. Jacobi's iterative

C. Gauss Jorden

D. Gauss elimination

Q.8 . .

The discretized form of 2D Poisson PDE when central difference approximation is applied on both derivatives with uniform step size h=0.5 in both directions is .....

A.

B. $4w_{i,j} - (w_{i-1,j} + w_{i+1,j}) - (w_{i,j-1} + w_{i,j+1}) = -\frac{1}{4} f_{i,j}$

C. $4w_{i,j} - (w_{i-1,j} + w_{i+1,j}) - (w_{i,j-1} + w_{i,j+1}) = -f_{i,j}$

D. $4w_{i,j} - (w_{i-1,j} + w_{i+1,j}) - (w_{i,j-1} + w_{i,j+1}) = \frac{1}{4} f_{i,j}$

$4w_{i,j} - (w_{i-1,j} + w_{i+1,j}) - (w_{i,j-1} + w_{i,j+1}) = -4 f_{i,j}$

Q.9 The Poisson Differential Equation \frac{\partial ^ 2 u}{\partial x ^2} + \frac{\partial ^ 2 u}{\partial y ^2} = f(x,y) becomes Laplace Differential Equation when

A. f(x,y)=xe^y

B. f(x,y)=0

C. f(x,y)=1

D. f(x,y)=e^x

Q.10 The Partial Differential Equation \frac{\partial ^ 2 u}{\partial x ^2} + \frac{\partial ^ 2 u}{\partial y ^2} = f(x,y) is

A. Poisson Differential Equation

B. Cauchy-Euler Differential Equation

C. Laplace Differential Equation

D. Legendre Differential Equation